

Kelly Criterion Derivation:

$w_0 \rightarrow$ init capital $b \rightarrow$ percent return (excluding init)
(return if we win)

$f \rightarrow$ optimal fraction to bet

$$\begin{aligned} \text{If won: } w_1 &= (1-f)w_0 + fw_0 + fbw_0 \quad \text{If lost: } \underline{w_1 = (1-f)w_0} \\ &= w_0 - f\cancel{w_0} + f\cancel{w_0} + fbw_0 \\ &= \underline{w_0(1-f)} \end{aligned}$$

$w = \# \text{ wins} \quad L = \# \text{ losses}$

$$w_n = w_0(1+bf)^w (1-f)^L$$

$$\ln\left(\frac{w_n}{w_0}\right) = w \ln(1+bf) + L \ln(1-f)$$

$\uparrow$
growth of wealth

$$\lim_{n \rightarrow \infty} \ln\left(\frac{w_n}{w_0}\right) = \overset{p}{\frac{w}{n}} \ln(1+bf) + \overset{q}{\frac{L}{n}} \ln(1-f)$$

$$g(f) = p \ln(1+bf) + q \ln(1-f)$$

maximize $g(f)$:

$$0 = \frac{pb}{1+bf} - \frac{q}{1-f} \Rightarrow \frac{q}{1-f} = \frac{pb}{1+bf}$$

$$\Rightarrow q + qbf = pb - pbf \Rightarrow pb - q = f b (q + p)$$

$$\Rightarrow \boxed{f = \frac{pb - q}{b}} \quad \square.$$